AP PRECALCULUS

Unit 1 • Worksheet 1.2C • Answer key

Worksheet 1.2C • Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$V(t)$ is the volume of water in a tank, in liters, t minutes after a valve opens. What are the units of the average rate of change of V?

(A) minutes per liter

(B) liters

(C) liters per minute

(D) liters times minutes

Solution

Answer: (C). Output units per input unit.

Question 2

NO CALCULATOR

The population of a town was 12,400 in 2010 and 15,100 in 2019. Which is the average rate of change of the population over this period?

(A) 270 people per year

(B) 300 people per year

(C) 2700 people per year

(D) 337.5 people per year

Solution

Answer: (B). $\frac{15100-12400}{9} = \frac{2700}{9} = 300$.

Question 3

NO CALCULATOR

A car's distance from home is $D(t)$ km, t hours after noon. $D(1) = 80$ and $D(3) = 20$. Which is the best interpretation of $\frac{D(3)-D(1)}{3-1}$?

(A) The car traveled 30 km.

(B) Between 1 p.m. and 3 p.m., the car's distance from home decreased on average by 30 km per hour.

(C) At 3 p.m. the car was moving at 30 km per hour.

(D) The car was 30 km from home at 2 p.m.

Solution

Answer: (B). The value is -30 km/h: an average rate over the interval, not a value at an instant.

Question 4 (Free response) NO CALCULATOR

The temperature T (in °F) in a city is recorded at times t (hours after midnight).

t	0	4	8	12	16
$T(t)$	52	48	57	71	66

(a) Find the average rate of change of T on $[4, 12]$. Interpret it in context with units.
(b) On which interval of length 4 did the temperature increase the fastest on average?
(c) Find $\frac{T(16)-T(0)}{16-0}$ and explain why it does not describe how the temperature changed during the day.

Solution
(a) $\frac{71-48}{8} = 2.875$. From 4 a.m. to noon, the temperature rose on average by 2.875 °F per hour.
(b) Rates: $[0,4]$: -1; $[4,8]$: 2.25; $[8,12]$: 3.5; $[12,16]$: -1.25. Fastest increase on $[8,12]$.
(c) $\frac{66-52}{16} = 0.875$ °F per hour. It hides the drop in the early morning and the drop in the afternoon; an average over a long interval can miss changes inside it.